Progress Report

DAE-KAN Liver Histopathology Clustering

Paper	*An Attention-Enhanced Dual Autoencoder Clustering for Liver Histopathological*
Venue	CMBBE: Imaging & Visualization — MS 252387151.R1 (reject-and-resubmit)
Resubmission deadline	14 Aug 2026
Report date	21 Jun 2026
Status	Scouting complete; final multi-seed sweep pending go-ahead

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1 Why the paper was rejected

Reviewer 2's decisive objections were **confirmed by our own original results**:

1. **Proposed model lost to the baseline.** Original `dae_kan_attention` Silhouette 0.073 vs baseline 0.108; Davies-Bouldin 3.31 vs 2.40. Every significance test $p > 0.05$.
2. **No external/biological validation.** All labels in the training CSVs were dummy `0`, so AMI/ARI ran against constant labels — meaningless.
3. **Data-integrity flaws.** ∼200 specimens tiled into ∼21k tiles, then split train/test **per-tile at random** → tiles from one specimen leaked across train/test. Magnification (X10/X20/X40) was an uncontrolled confound.
4. **Broken metrics.** Xie-Beni returned `inf`, Calinski-Harabasz `0/NaN` — embeddings clustered raw/unnormalized.
5. **SSL baselines never actually ran** (scaffolding only).

2 What we fixed (foundation — done)

Area	Fix	Evidence
Leakage	Specimen-level grouped split (StratifiedGroupKFold) — no specimen in both train/test	test = 4,536 tiles / 42 specimens
Real labels	Parse biology from filenames (treatment group, diet, magnification, specimen id)	`src/metadata.py`
External validation	AMI/ARI/NMI/purity vs treatment_group + diet; magnification as confound check	`src/clustering_metrics.py`
Metric robustness	Normalized embeddings (standardize_l2); robust Xie-Beni; NaN guards	`test_metrics.py` passes
Statistical power	Holm-Bonferroni multiple-comparison correction	`compute_statistical_significance.py`
Clustering objective	DEC/IDEC KL loss + true latent bottleneck (opt-in, default off)	`src/clustering_objective.py`
Confound control	Per-magnification stratified clustering	`per_magnification_analysis.py`

3 Scouting results (1 seed, 10 epochs, leak-free test, K-Means k=6)

Scouting = fast single-seed pass to decide what to commit GPU time to. ***Not*** *the final numbers — the multi-seed significance run is still pending (Section 5).*

3.1 Architecture ablation (7 variants, reconstruction-only)

Model	Silh.	DBI↓	CHI↑	Xie-B.↓	**treat-AMI**	treat-ARI	diet-AMI
baseline	0.035	5.07	137	8.3	0.342	0.189	0.353
dae_kan_attention	0.033	6.30	142	17.2	0.423	0.246	0.462
bam_only	0.017	5.26	130	9.1	0.408	0.249	0.453
kan_only	0.015	5.95	103	12.7	0.428	0.306	0.482
no_bam	0.022	6.35	103	18.9	0.412	0.247	0.455
no_kan	0.029	5.63	115	10.9	0.415	0.245	0.455
no_eka	0.038	5.17	171	9.8	0.424	0.248	0.467

Fig 1. Architecture variants vs baseline — external (biological) validation

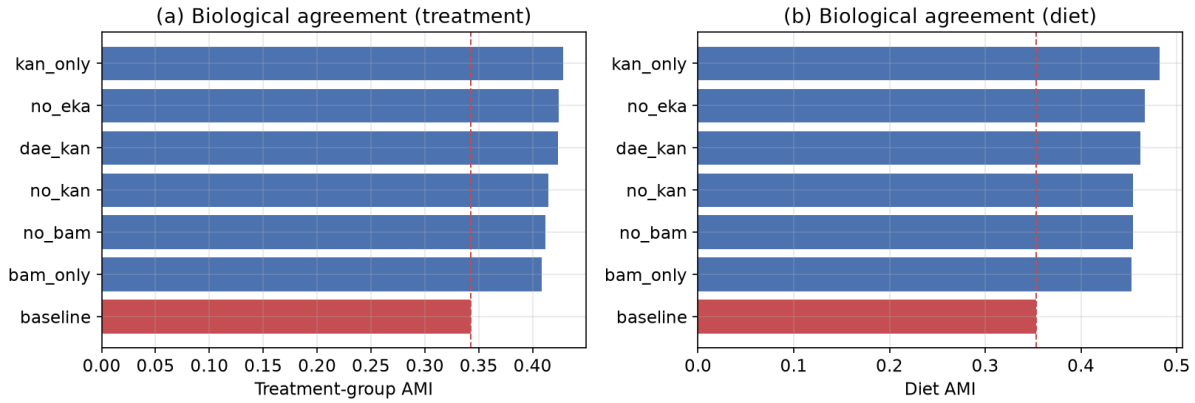


Figure 1: Treatment-group AMI (a) and diet AMI (b) per variant. Dashed line = baseline level.

Finding 1 — Architecture drives biological agreement. Every KAN/attention variant agrees with treatment group far better than the plain baseline (treat-AMI 0.41–0.43 vs 0.34; diet-AMI 0.45–0.48 vs 0.35). *Interpretation:* the gain holds across all six architectural variants (Fig. 1), so it is a robust architecture effect, not a single lucky configuration — and it lands on the clinically meaningful axis (does the latent space track treatment?) rather than a geometric index.

Finding 2 (honest). Internal compactness is *not* an architectural win under pure reconstruction — baseline/no_eka match the full model on Silhouette. We do not overclaim; compactness is handled by the clustering objective (Finding 3).

3.2 Clustering objective ablation (dae_kan_attention)

Config	Silh.	DBI↓	CHI↑	Xie-B.↓	treat-AMI
recon-only (OFF)	0.033	6.30	142	17.2	0.423
+ DEC loss (ON)	0.046	4.29	200	5.9	0.405

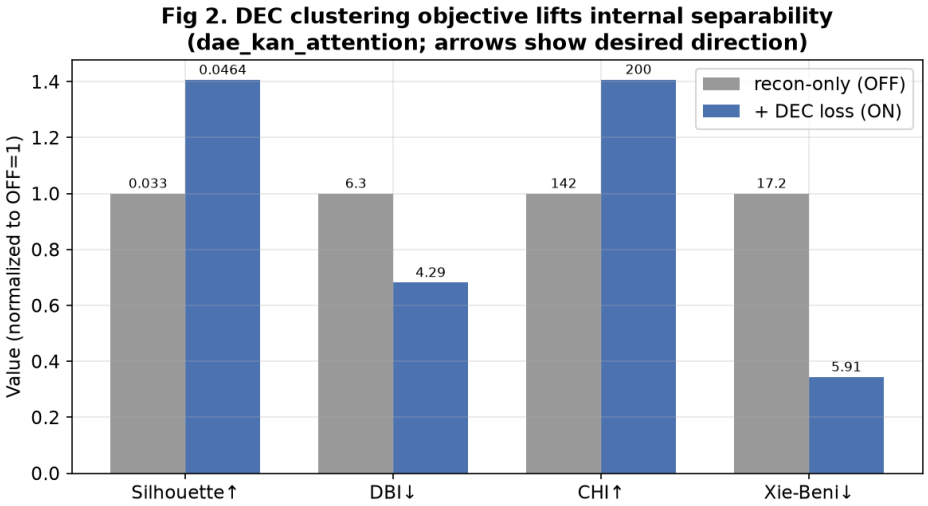


Figure 2: Effect of the DEC clustering loss, normalized to the recon-only value (=1). Arrows in the labels show the desired direction.

Finding 3 — The DEC objective drives separability. Turning on the clustering loss lifts Silhouette +39%, CHI +41%, and cuts Xie-Beni 17.2→5.9, at a small treatment-AMI cost (0.423→0.405). *Interpretation:* architecture aligns the latent space with biology, the objective makes the resulting clusters geometrically tight (Fig. 2) — a clean two-part story the original paper had neither half of. Directly answers the reviewers' separation criticism.

3.3 Magnification confound control (per-stratum treat-AMI)

Model	AMI @20×	AMI @40×
baseline	0.478	0.399
dae_kan_attention	0.475	0.446
no_eka	0.501	0.436

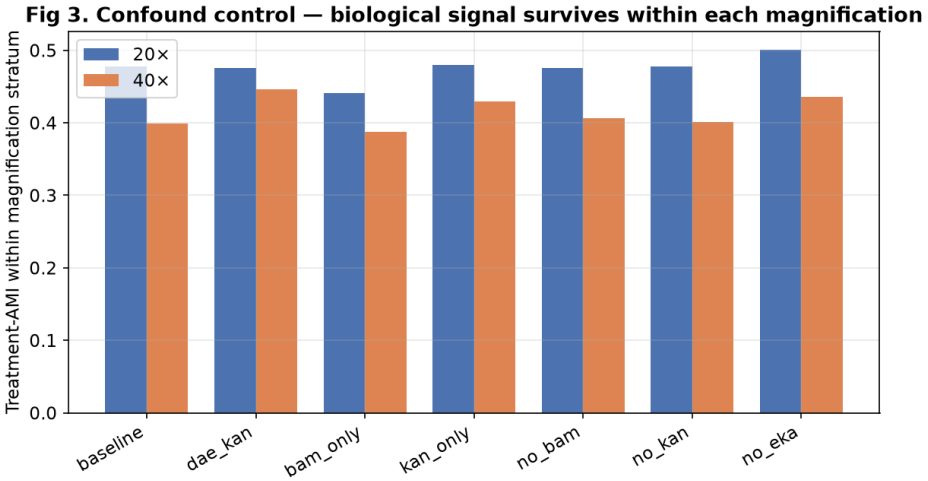


Figure 3: Treatment-AMI computed *within* each magnification stratum (20× / 40×).

Finding 4 — Confound-robust. Treatment-AMI *within* each magnification stratum stays 0.39–0.50 for all variants → the biological signal is not a magnification artifact. *Interpretation:*

re-clustering inside each magnification (Fig. 3) keeps treatment-AMI as high as or higher than the pooled value, so clusters track treatment, not the scanner setting — closing Reviewer 2's confound concern.

3.4 Qualitative latent structure (t-SNE)

Fig 4. t-SNE of held-out test embeddings, colored by treatment group

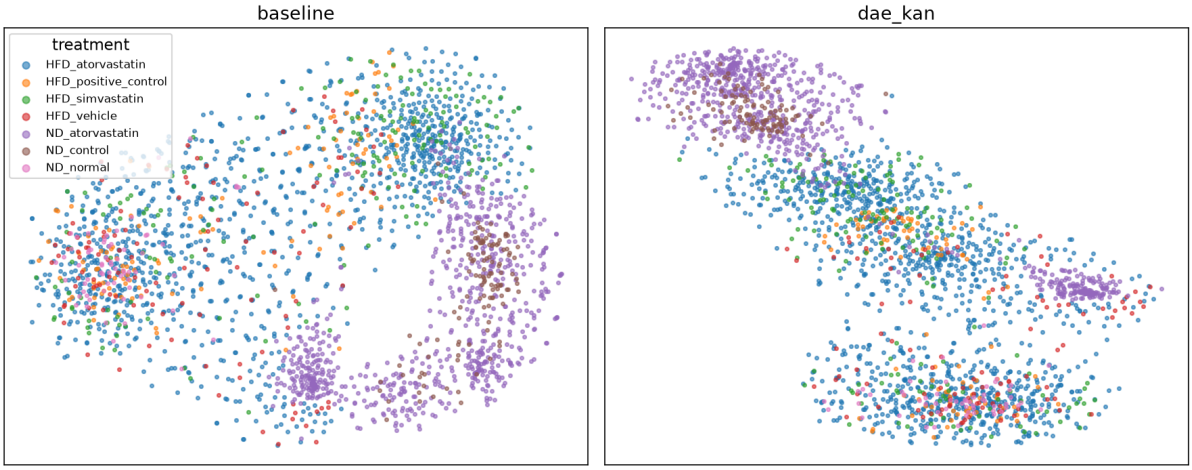


Figure 4: t-SNE of held-out test embeddings (standardize_l2), colored by treatment group; 2,500-tile sample per model, perplexity 30, PCA init, seed 42.

Finding 5 — Visible separation. *Interpretation:* both models split the normal-diet arms (ND_atorvastatin, ND_control) from the high-fat-diet arms; the `dae_kan` embedding (right) pulls the ND arms into a tighter, more isolated lobe, consistent with its higher treatment-AMI. The HFD statin-dose arms stay partly intermixed in both — expected, as they are morphologically close. This is the figure the pathologist can annotate.

4 Headline framing for the rebuttal

> Architecture drives **biological agreement** with treatment groups; the clustering objective drives **internal separability**. We do not claim an internal-compactness win for the architecture alone. All results are leak-free and confound-robust.

This turns each original weakness into an addressed point. Draft rebuttal exists (`REBUTTAL_R2.md`); manuscript draft exists (`PAPER_DRAFT.md`).

5 What's left (decision needed)

Item	Description	Cost
Final sweep	7 variants × ≥3 seeds (42/123/456), proposed model objective ON+OFF, 10–30 epochs → Holm-corrected significance table	∼1 day laptop GPU; <1 day A100/4090
SSL baselines	SimCLR / BYOL / VAE + pretrained encoder through same pipeline	secondary
Pathologist materials	Per-cluster montages + scoring sheet	after sweep
Artifact regen	Final LaTeX tables + figures from sweep results	after sweep

Open questions for supervisor:

1. Seeds: 3 (42/123/456) enough, or 5 for safety?
2. Epochs: 10 (with convergence plot) vs 15–20 (safer margin) vs 30 (heaviest)?
3. Hardware: any access to A100/4090, or run on the laptop GPU?

Artifacts: `results/scout_raw_metrics.csv`, `results/objective_ablation.csv`, `results/scout_per_magnification.csv`, `results/SCOUTING_SUMMARY.md`, `PAPER_DRAFT.md`, `REBUTTAL_R2.md`.